Parameter-Efficient Reinforcement Learning via Dynamic Sparse Training: Enhancing A2C with DAPD across Classic Control and Atari Environments

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Abstract

We explore parameter-efficient reinforcement learning by integrating Dynamic Sparse Training with Pathways (DAPD) into the Advantage Actor-Critic (A2C) framework. To understand whether DAPD's limitations are algorithm-specific or more general, we compare it against both dense A2C and Proximal Policy Optimization (PPO) across three environments: LunarLander-v2, Breakout, and Pong. Our experiments show that while DAPD reduces training time significantly, it introduces learning instability and degrades performance, especially in control tasks. PPO consistently achieves higher rewards and greater stability. These results highlight a key trade-off in sparse training: improved computational efficiency at the cost of reduced convergence reliability.

1 Introduction

Deep reinforcement learning (RL) has shown remarkable performance in domains ranging from continuous control to visual decision-making. However, this often comes at the cost of substantial computational resources, particularly due to the size and complexity of neural networks used in RL agents. In real-world applications, especially those involving deployment on edge devices or embedded systems, such requirements can be prohibitive.

To address these challenges, we study parameter-efficient RL through Dynamic Sparse Training with Pathways (DAPD) [Samin Yeasar Arnob and Precup, 2024], a training paradigm that maintains a sparse subset of active connections throughout learning. This approach builds on recent advances in dynamic sparsity and aims to reduce the parameter footprint without sacrificing performance.

Compared to our original proposal, our final study departs in two key ways. First, we initially planned to explore a variant we called Double A2C, but early results indicated that its performance was not meaningfully different from standard A2C, and therefore not analytically interesting. Second, although we successfully implemented the PoPS method, we did not have sufficient time to conduct a thorough analysis to draw meaningful comparisons with other baselines. As a result, our focus shifted to fully exploring the impact of DAPD within a fixed experimental budget.

Our work investigates the application of DAPD in the Advantage Actor-Critic (A2C) algorithm and compares it to both dense A2C and Proximal Policy Optimization (PPO), a widely used high-performing baseline. This design allows us to disentangle the impact of sparsity from the characteristics of the underlying algorithm. We apply these methods across three RL environments: LunarLander-v2, Breakout, and Pong.

Our contributions include:

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- Evaluating the impact of DAPD on the A2C algorithm across both classic and visual reinforcement learning environments.
- Demonstrating the instability of sparse training in control tasks and its moderate effectiveness in visual domains.
- Providing empirical insights into how DAPD compares with dense baselines such as A2C and PPO [Xin Liu and Razaviyayn, 2021].

2 Background and Related Work

A2C and PPO are two widely used actor-critic algorithms in reinforcement learning. A2C provides synchronous policy updates with reduced variance, while PPO improves training stability through clipped objective functions. Both have demonstrated strong performance across a range of tasks, including classic control problems and high-dimensional visual environments.

In this work, we focus on three representative environments: **LunarLander-v2**, a low-dimensional control task from the Box2D suite; and two Atari games, **Breakout** and **Pong**, which pose greater challenges due to their pixel-based observations and sparse rewards. This selection enables us to assess algorithmic behavior under varying complexity and observation modalities.

Dynamic Sparse Training (DST), including variants such as RigL and DAPD, has achieved strong compression–performance trade-offs in supervised learning [Samin Yeasar Arnob and Precup, 2024]. These methods maintain a sparse parameter mask throughout training, periodically adapting it to shift model capacity toward more salient weights.

3 Methodology

We evaluate three agent configurations: standard A2C, A2C with Dynamic Sparse Training (DAPD), and PPO. These setups enable us to explore performance and efficiency trade-offs in reinforcement learning.

A2C. The Advantage Actor-Critic algorithm includes a policy (actor) and value (critic) network trained using advantage-weighted policy gradients. Both networks share a simple two-layer feedforward architecture.

PPO. PPO uses the same architecture but improves training stability through a clipped objective, repeating updates over mini-batches to enhance sample efficiency. It serves as a strong dense baseline [Xin Liu and Razaviyayn, 2021].

A2C+DAPD. In this configuration, A2C is augmented with dynamic sparsity: a binary mask determines active weights, and prune–regrow updates periodically reallocate capacity. Sparsity remains fixed while the mask adapts to training needs.

We conduct experiments in three environments with different characteristics:

- **LunarLander-v2:** A low-dimensional control task involving a lunar module attempting to land safely.
- **Breakout:** A vision-based Atari game requiring spatial reasoning.
- **Pong:** Another Atari game focused on visual and temporal coordination.

All agents are trained for 10,000 episodes in each environment. We record episodic rewards, compute the average reward over the last 100 episodes, and measure total wall-clock training time. We also generate training reward curves for direct comparison of learning stability, convergence behavior, and final performance.

4 Experiments

We evaluate the performance of three reinforcement learning agents—A2C, A2C+DAPD, and PPO—across three benchmark environments: LunarLander-v2, Breakout, and Pong. Our analysis focuses on learning behavior, computational efficiency, and stability under different sparsity regimes.

All models are trained using PyTorch on a single NVIDIA RTX 2080 Ti GPU with a fixed random seed of 1. The policy and value networks for each agent share the same architecture, consisting of two hidden layers with 64 ReLU units each. Optimization is conducted using the Adam optimizer with a learning rate of 1×10^{-4} for A2C and DAPD variants. We use a discount factor $\gamma = 0.99$, GAE with $\lambda = 0.95$, and a rollout length of 128 steps per update.

For the DAPD variant, we enforce a sparsity level of 90% by retaining only 10% of the weights. Masks are updated every 1,000 training steps on LunarLander and every 100,000 steps on Breakout and Pong, to balance the frequency of structural updates with computational overhead. All environments are trained for a total of 10,000 episodes, and smoothed reward curves are reported using a moving average over 100 episodes.

In terms of wall-clock training time, PPO converges fastest on LunarLander-v2 (≈14,000 s), followed by A2C (≈25,000 s), while A2C+DAPD converges more quickly (≈8,000 s) but exhibits greater variance. Similar patterns are observed in Breakout (PPO: 10,000 s; A2C: 11,000 s; DAPD: 7,000 s) and Pong (PPO: 16,000 s; A2C: 21,000 s; DAPD: 14,000 s), suggesting that sparse training can accelerate learning at the cost of reduced stability.

Our full implementation, including training scripts, configurations, and logging utilities, is publicly available at:
`https://github.com/yyn-anson/COMP579_FinalProject`

This repository supports reproducibility and provides utilities for switching between dense and sparse modes, adjusting pruning schedules, and logging reward/time metrics automatically.

5 Conclusion and Future Work

This work explored the integration of Dynamic Active Path Discovery (DAPD) into the A2C framework, evaluating its impact on both classic control and visual reinforcement learning tasks. DAPD showed promise in accelerating early-stage training through enforced sparsity but struggled with stability and final performance, particularly in visually rich environments.

While PPO remained robust across settings, the difficulties faced by DAPD suggest that sparsity introduces general challenges to policy gradient methods. This calls for improved sparsity-aware optimization strategies that can preserve learning stability.

Although we successfully implemented the Policy Pruning and Shrinking (PoPS) method, we were unable to complete its evaluation due to the extended training time required across multiple environments. Future work will include a full empirical analysis of PoPS under different pruning schedules and a direct comparison with DAPD.

We also plan to revisit other variants like Double A2C, and explore structured sparsity, adaptive pruning, and hybrid models. Long-term goals include testing these techniques in real-time deployment scenarios on resource-limited hardware.

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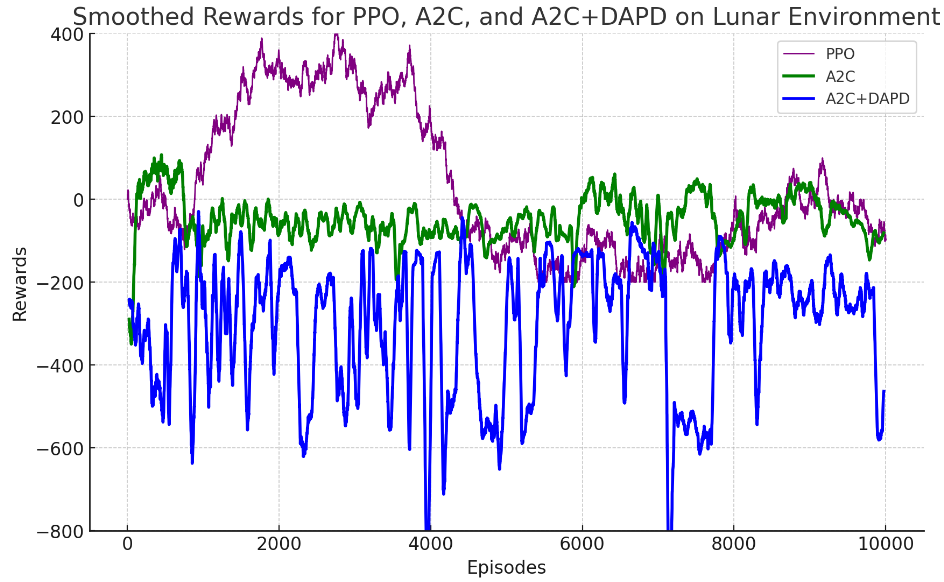
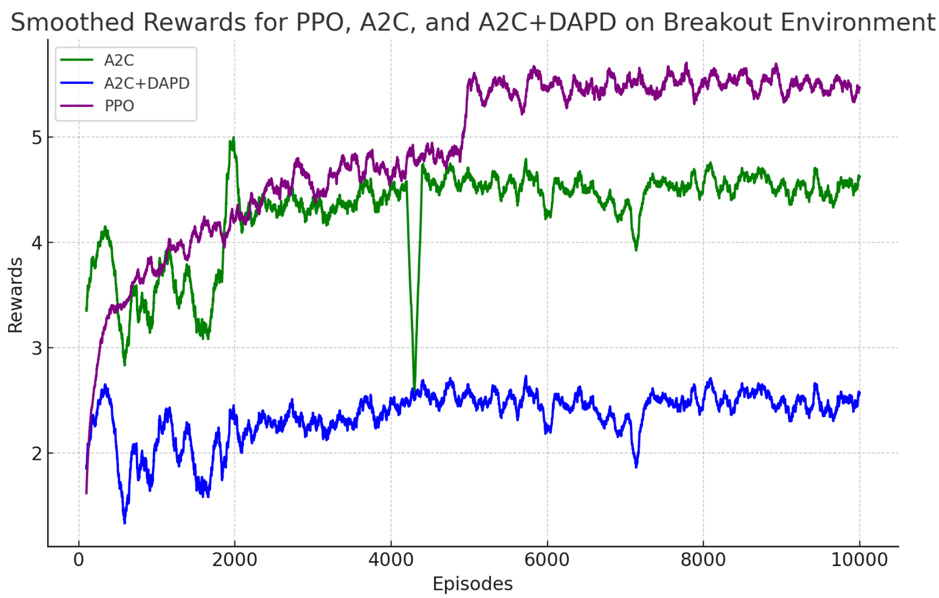
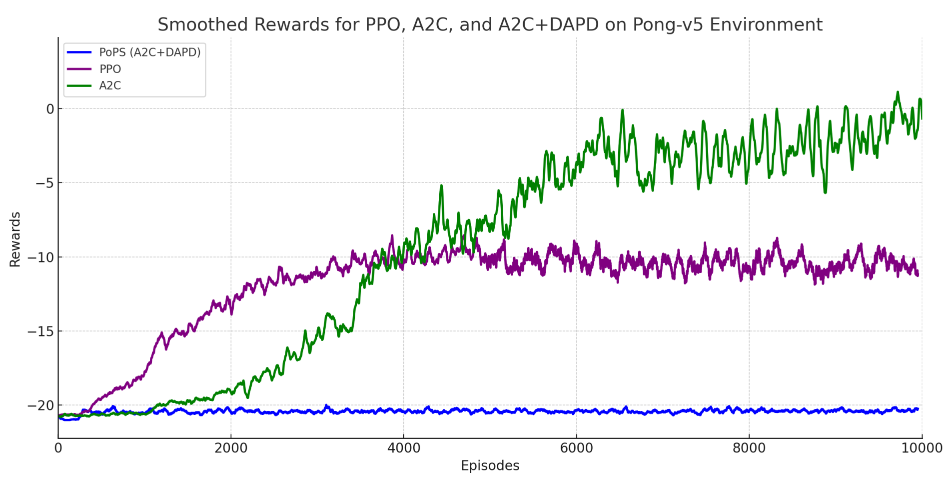


(a) LunarLander-v2



(b) Breakout



(c) Pong

Figure 1: Smoothed rewards for PPO, A2C, and A2C+DAPD across three environments.